1.3 Hazards

1. The Concept of Hazard

Natural hazard: A natural event or process that has the potential to cause loss of life, injury, or damage to property, infrastructure, and livelihoods. A hazard only becomes a *disaster* when it significantly impacts a vulnerable population. The same magnitude event can be a minor hazard in one location and a catastrophic disaster in another, depending on vulnerability and capacity to cope.

Disaster: A hazard event causing significant damage and disruption that exceeds a community's ability to cope using its own resources. The UN threshold: 10+ deaths, 100+ people affected, a state of emergency declared, or international assistance called for. Between 2000 and 2019, there were 7,348 recorded disasters worldwide, killing 1.23 million people and affecting 4.2 billion (UNDRR).

The **hazard risk equation**:

$$\text{Risk} = (\text{Hazard} \times \text{Vulnerability}) \ / \ \text{Capacity to Cope}$$

Key concepts:

- **Vulnerability** — the susceptibility of a population to the impacts of a hazard. Influenced by poverty, population density, building quality, governance, education, social inequality, age structure, and marginalisation. LICs are disproportionately vulnerable: 95% of disaster deaths occur in developing countries.
- **Exposure** — the degree to which people and assets are located in hazardous areas. Increasing due to population growth, urbanisation, and coastal settlement.
- **Resilience** — the ability of a community to absorb the impact, adapt, and recover. Built through preparedness, early warning systems, insurance, social networks, and institutional capacity.

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- **Capacity to cope** — resources, knowledge, technology, and infrastructure available for preparation, response, and recovery. HICs generally have greater capacity through emergency services, building regulations, insurance, and financial reserves.

Hazard Perception and Response

People perceive hazard risk differently depending on experience, wealth, education, personality, and cultural/religious beliefs. Three common responses to hazards:

- **Fatalism** — accepting hazards as inevitable, "acts of God", or fate, doing little to prepare. More common in deeply religious or disadvantaged communities with limited resources.
- **Adaptation** — adjusting behaviour and modifying the environment to reduce risk (e.g. building earthquake-resistant structures, crop diversification, land-use zoning). The most proactive and effective response.
- **Fear/avoidance** — an extreme response that may lead to out-migration from hazardous areas or excessive anxiety. Can lead to NIMBYism (resisting development in hazardous areas).

Perception is influenced by the **recency effect** (recent events seem more likely), **optimism bias** (believing "it won't happen to me"), and **magnitude-frequency relationship** (high-magnitude events are rare, so people underestimate their likelihood).

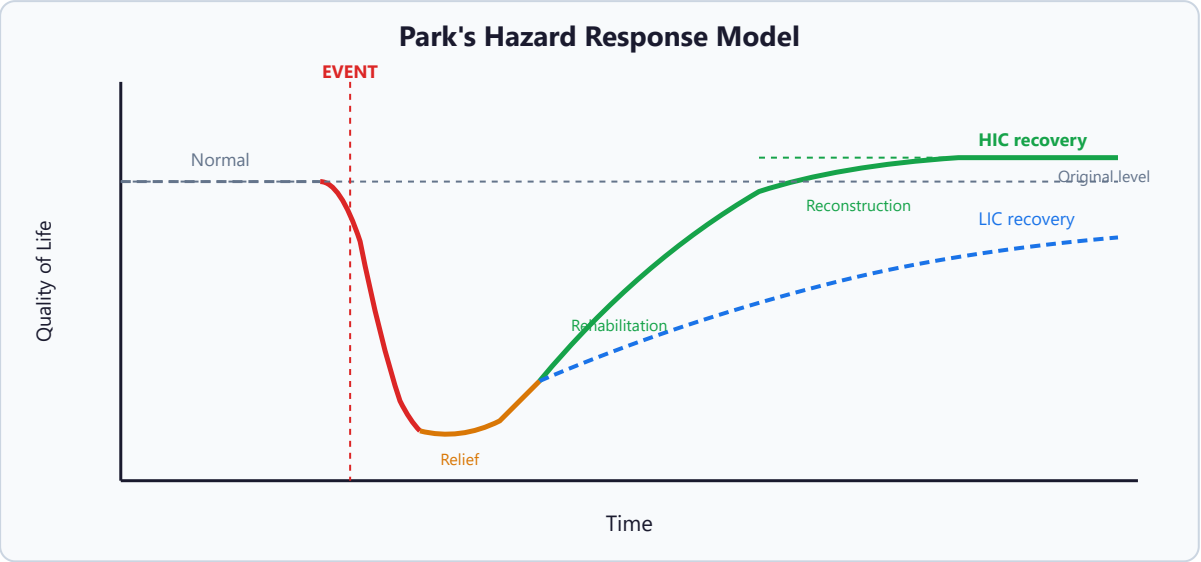
Park's Hazard Response Model

This model describes the stages a community passes through after a disaster event:

1. **Pre-disaster** — normal functioning, ideally with preparedness measures in place.
2. **The event** — the hazard strikes, causing immediate disruption and decline in quality of life.
3. **Search and rescue** — emergency response phase (hours to days). Focus on saving lives, providing immediate shelter and medical care.
4. **Relief** — short-term humanitarian aid (days to weeks). Food, water, temporary shelter, sanitation.
5. **Rehabilitation** — restoring services, infrastructure, and economic activity (weeks to months).
6. **Reconstruction** — long-term rebuilding, ideally to a higher standard ("build back better") to reduce future vulnerability (months to years).

The depth of the decline and speed of recovery depend on: event magnitude, vulnerability, economic resources, governance quality, international aid, and pre-existing preparedness. In HICs, recovery is

typically faster but costs are higher in monetary terms. In LICs, recovery may take decades and development gains may be permanently lost.



The Hazard Management Cycle

Effective hazard management is a continuous cycle of four stages:

- **Mitigation** — reducing the severity of hazard impacts through structural measures (building codes, flood defences, land-use zoning) and non-structural measures (insurance, education, legislation). The most cost-effective stage.
- **Preparedness** — planning for a disaster: emergency plans, warning systems, evacuation drills, stockpiling supplies, training emergency services.
- **Response** — actions taken immediately after a disaster: search and rescue, emergency shelter, medical treatment, humanitarian aid distribution.
- **Recovery** — restoring normality: rebuilding infrastructure, economic recovery, psychological support, lessons learned to improve future mitigation.

2. Plate Tectonics

Plate tectonics: The theory that the Earth's lithosphere is divided into ~15 major rigid plates (and several minor ones) that move relative to one another on the semi-plastic asthenosphere. Plate movements are driven by convection currents in the mantle, slab pull, and ridge push. The theory,

developed in the 1960s, unified earlier ideas of continental drift (Wegener, 1912) and sea-floor spreading (Hess, 1962).

Earth's Internal Structure

- **Inner core** — solid iron and nickel, ~5,500°C, ~1,220 km radius. Solid despite extreme temperature because of immense pressure.
- **Outer core** — liquid iron and nickel, generates Earth's magnetic field through convection, ~2,200 km thick.
- **Mantle** — silicate rock, ~2,900 km thick. Upper mantle includes the rigid lithosphere (crust + uppermost mantle, ~100 km thick) and the plastic asthenosphere (partially molten, allowing plates to move).
- **Crust** — thin outer layer. *Oceanic crust*: thin (5–10 km), dense (~3.0 g/cm^3), basaltic, geologically young (<200 million years). *Continental crust*: thick (25–70 km), less dense (~2.7 g/cm^3), granitic, geologically old (up to 4 billion years).

Evidence for Plate Tectonics

- **Continental fit** — jigsaw-like fit of South America and Africa (first noted by Wegener, 1912).
- **Geological match** — identical rock types, mountain chains, and fossil distributions across continents now separated by oceans (e.g. Glossopteris fern fossils on South America, Africa, India, Australia, and Antarctica).
- **Palaeomagnetism** — symmetrical magnetic striping on the ocean floor either side of mid-ocean ridges, recording periodic magnetic field reversals as new crust formed.
- **Sea-floor spreading** — younger rocks near ridges, progressively older rocks further away (Harry Hess, 1960s). Confirmed by deep-sea drilling.
- **GPS measurements** — direct measurement of plate movements: North Atlantic widening at ~2.5 cm/yr, Pacific plate moving NW at ~7 cm/yr.
- **Distribution of earthquakes and volcanoes** — concentrated along plate boundaries, confirming their locations.

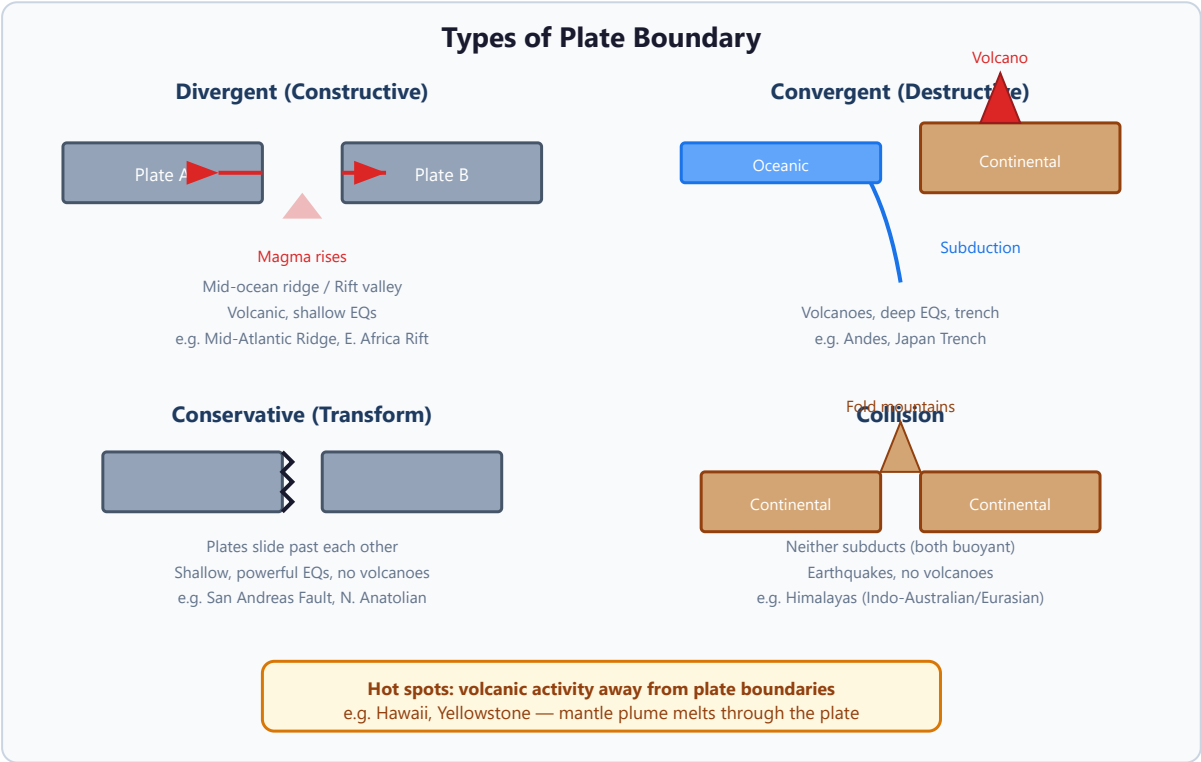
Driving Forces of Plate Movement

- **Mantle convection** — heat from radioactive decay in the core creates convection currents in the mantle. Rising magma at ridges pushes plates apart; sinking material at subduction zones pulls plates

down.

- **Slab pull** — the weight of a cold, dense subducting plate (oceanic lithosphere) pulls the rest of the plate with it. Now considered the dominant driving force, responsible for ~90% of plate motion.
- **Ridge push** — gravity causes newly formed, elevated crust at mid-ocean ridges to slide away from the ridge axis.
- **Basal drag** — friction between the convecting mantle and the overlying plate may drag the plate along, though this is debated.

Types of Plate Boundary

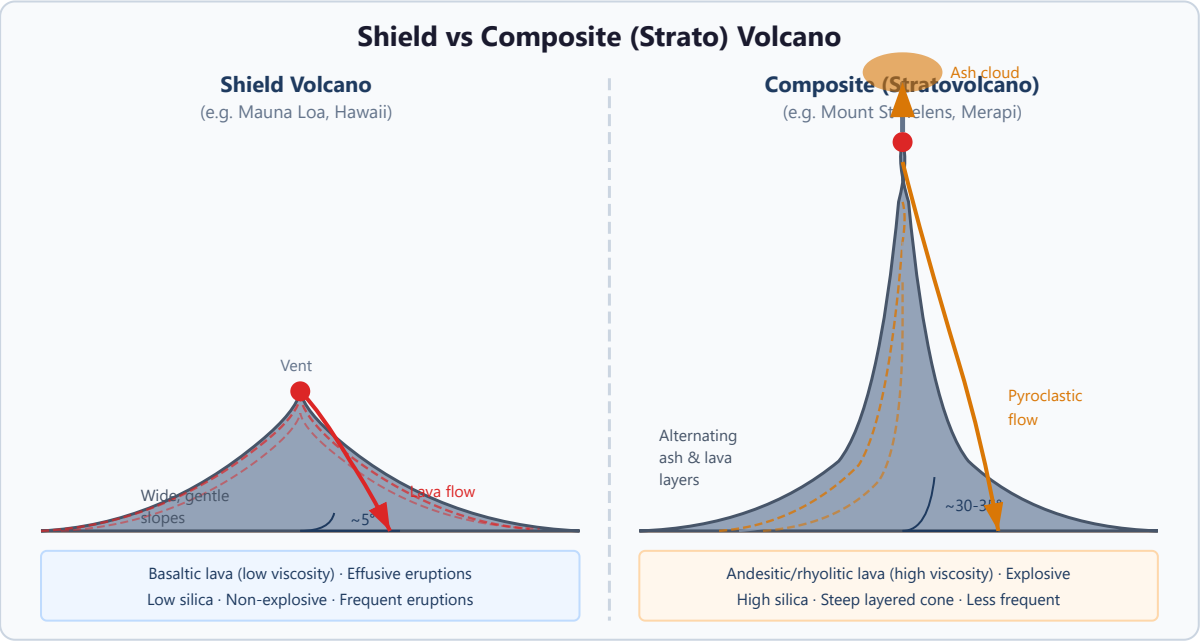


3. Volcanic Hazards

Volcano: An opening in the Earth's crust through which magma, ash, and gases are ejected. Volcanic eruptions range from gentle effusive flows (basaltic lava, low silica, low viscosity) to violent explosive eruptions (rhyolitic magma, high silica, high viscosity). Approximately 1,500 potentially active volcanoes exist worldwide, with ~50–70 erupting each year.

Types of Volcano

Type	Shape	Eruption Style	Lava/Magma	Size	Example
Shield	Wide, gently sloping (2–10°)	Effusive, frequent, non-explosive	Basaltic (low viscosity, 1,000–1,200°C)	Very large (can be 100+ km diameter)	Mauna Loa, Hawaii (4,169 m)
Composite (stratovolcano)	Steep, conical (25–35°), layered ash and lava	Explosive, less frequent, violent	Andesitic/rhyolitic (high viscosity, 800–1,000°C)	Medium (2–3 km high)	Mount St Helens, Mount Merapi
Cinder cone	Small, steep-sided	Short-lived, moderate explosions	Basaltic to andesitic	Small (<300 m high)	Paricutin, Mexico
Caldera (supervolcano)	Large depression (10–100 km wide)	Catastrophic, extremely rare (every 50,000–100,000 years)	Rhyolitic (very high viscosity)	Enormous eruption volume	Yellowstone (USA), Toba (Indonesia)



Volcanic Hazards

- **Lava flows** — streams of molten rock. Basaltic flows can travel at 10–60 km/h, reaching 30+ km from the vent. Slow enough to evacuate from, but destroy everything in their path. Rhyolitic lava moves slowly (<1 km/h) due to high viscosity.
- **Pyroclastic flows** — superheated avalanches of gas, ash, and rock fragments travelling at 100–700 km/h, temperatures up to 1,000°C. Virtually impossible to escape. The most lethal volcanic hazard. Destroyed Pompeii (AD 79) and Saint-Pierre, Martinique (1902, 29,000 killed).
- **Lahars** — volcanic mudflows formed when ash mixes with water (rainfall, snowmelt, crater lake overflow). Can travel 50–300 km/h, reaching distances of 300+ km from the volcano. Nevado del Ruiz, Colombia (1985): lahars travelled 100 km and killed 23,000 in the town of Armero.
- **Tephra and ash fall** — rock fragments (tephra) and fine ash ejected into the atmosphere. Ash can collapse roofs (>10 cm accumulation), disrupt aviation (Eyjafjallajokull 2010 grounded 100,000 flights, cost airlines ~$1.7 billion), contaminate water, damage crops, and cause respiratory problems.
- **Volcanic gases** — CO_2, SO_2, H_2S, HF. Can cause acid rain, respiratory problems, and climate impacts. Major eruptions inject SO_2 into the stratosphere, forming sulphate aerosols that reflect sunlight and cause temporary global cooling (e.g. Mount Pinatubo 1991 cooled the planet by ~0.5°C for 2 years).
- **Tsunami** — volcanic eruptions on islands, submarine volcanoes, or caldera collapses can trigger devastating tsunamis (e.g. Krakatoa 1883: tsunamis up to 30 m killed 36,000).
- **Volcanic earthquakes** — seismic activity caused by magma movement beneath the volcano. Used for monitoring and prediction.

Case Study: Mount Merapi, Indonesia (2010)

Mount Merapi is one of the world's most active stratovolcanoes, located on a destructive boundary (Indo-Australian plate subducting beneath the Eurasian plate) in central Java. 25 million people live within 30 km. The October–November 2010 eruption (VEI 4) produced pyroclastic flows reaching 15+ km from the summit, ash columns to 18 km altitude, and lahars during the monsoon season. **Impacts:** 353 deaths, 350,000 displaced, 2,682 houses destroyed, ~$600 million economic damage, agricultural land buried under ash, livestock killed. **Response:** Indonesia's PVMBG (volcanology agency) raised the alert to maximum 4 days before the main eruption; 20,000 evacuated. The government established a 20 km exclusion zone. However, many residents returned prematurely to tend livestock and crops, increasing casualties. **Lessons:** improved monitoring since (real-time GPS, seismic sensors, webcams), updated hazard maps, permanent evacuation routes and shelters. The eruption highlighted tensions between scientific warnings and local livelihoods: people living on Merapi's fertile volcanic soils depend on farming and cannot easily relocate.

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Case Study: Eyjafjallajokull, Iceland (2010)

An eruption at the Eyjafjallajokull volcano (March–May 2010) produced enormous ash clouds that disrupted European airspace for 6 days in April 2010. Iceland sits on the Mid-Atlantic Ridge (divergent boundary) and a hot spot. The eruption was moderate (VEI 4) but the fine-grained ash, jet stream direction, and glacial meltwater interacting with magma (phreatomagmatic eruption) created exceptionally fine ash particles that spread across Northern Europe. **Impacts:** ~100,000 flights cancelled, 10 million passengers stranded, airline losses ~$1.7 billion, supply chain disruptions (flower exports from Kenya declined 50%, car plants in Germany halted), but no direct deaths. Iceland: 800 evacuated from flooding caused by glacial melt (jokulhlaup), farmland covered in ash. **Response:** European aviation authorities initially imposed a blanket flight ban, later revised to allow flights with defined ash concentration limits. Post-event: improved volcanic ash monitoring, revised aviation protocols, development of the International Airways Volcano Watch.

4. Seismic Hazards

Earthquake: A sudden release of energy in the Earth's crust or upper mantle, generating seismic waves. Most earthquakes occur at plate boundaries due to friction between moving plates. Energy builds up when plates lock, then is released suddenly when the stress overcomes friction. The point of energy release within the Earth is the **focus (hypocentre)**; the point on the surface directly above is the **epicentre**.

Measuring Earthquakes

Scale	What It Measures	Key Features
Moment Magnitude Scale (M_W)	Total energy released (seismic moment)	Logarithmic (each whole number = ~32x more energy). No upper limit. Replaced the Richter scale. Used by seismologists worldwide.
Modified Mercalli Scale (MMI)	Intensity of shaking and damage observed	Qualitative, I (not felt) to XII (total destruction). Varies with distance from epicentre, geology, building quality. Useful for comparing impacts.

Factors Affecting Earthquake Impact

- **Magnitude** — larger earthquakes release more energy and cause more widespread damage.
- **Depth of focus** — shallow-focus earthquakes (<70 km) are generally more destructive than deep-focus (>300 km) because energy is concentrated near the surface.
- **Distance from epicentre** — shaking intensity decreases with distance.
- **Population density** — earthquakes in remote areas may cause few casualties; the same magnitude in a city can be catastrophic.
- **Building quality** — earthquake-resistant buildings (steel frame, base isolation, cross-bracing) dramatically reduce collapse risk. Many deaths result from building collapse, not the earthquake itself.
- **Ground conditions** — liquefaction (saturated, loose sediment behaves like liquid during shaking) can cause buildings to sink. The 2011 Christchurch earthquake caused extensive liquefaction in areas built on river silt.
- **Time of day** — earthquakes at night cause more casualties (people sleeping in buildings).
- **Secondary hazards** — tsunamis, landslides, fires (ruptured gas mains), dam failure, disease outbreaks.
- **Level of preparedness** — early warning systems, building codes, emergency services, education, and drills.

Case Study: Nepal Earthquake (25 April 2015) — LIC

A magnitude 7.8 earthquake struck central Nepal at a shallow depth of ~15 km, on the collision boundary between the Indian and Eurasian plates (the Himalayas are still rising at ~5 mm/yr). **Primary impacts:** 8,964 deaths, 22,309 injured, 604,930 houses destroyed, 288,856 damaged, ~$7 billion total damage (50% of Nepal's GDP). Historic temples in Kathmandu's Durbar Square collapsed. Thousands of landslides blocked roads. **Secondary impacts:** avalanche on Mount Everest killed 22, aftershock (M 7.3, 12 May) caused further destruction, 3.5 million left homeless, cholera and disease risk from contaminated water, schools and hospitals destroyed. **Response:** international aid ($4.1 billion pledged), Indian and Chinese military deployed, Nepalese army led rescue. However, aid was slow to reach remote mountain villages due to damaged roads and rugged terrain. **Recovery:** slow — 3 years later, only 5% of destroyed homes had been rebuilt. Weak governance, corruption, and political instability hampered reconstruction. Nepal's vulnerability: poverty (25% below poverty line), weak building codes, rapid unplanned urbanisation, limited emergency services.

Case Study: Tohoku Earthquake and Tsunami, Japan (11 March 2011) — HIC

A magnitude 9.0 megathrust earthquake struck offshore NE Japan on the destructive boundary where the Pacific plate subducts beneath the North American plate. Focus at 32 km depth. **Earthquake impacts:** ground shaking lasted 6 minutes (one of longest recorded), extensive damage to infrastructure despite Japan's world-leading earthquake engineering. **Tsunami:** waves up to 40 m high struck the coast within 30 minutes, overwhelming sea walls designed for 5–10 m waves. The tsunami penetrated up to 10 km inland. **Total impacts:** 19,747 deaths (93% from drowning in tsunami), 6,242 injured, 470,000 evacuated, 127,000+ buildings destroyed. Fukushima Daiichi nuclear power plant lost cooling systems, causing 3 reactor meltdowns and the world's worst nuclear disaster since Chernobyl (160,000 evacuated from the exclusion zone). Economic cost: ~$235 billion (costliest natural disaster in history). **Response:** Japan's early warning system gave 8–30 minutes' warning after the earthquake; many coastal residents evacuated. Self-Defence Forces deployed within hours. 130+ countries offered aid. **Recovery:** Japan's strong governance, high GDP, and insurance coverage enabled rapid infrastructure reconstruction, but emotional and community recovery was slower. 10 years later, some coastal towns had not been rebuilt. New 12.5 m sea walls constructed, revised tsunami warning systems, nuclear safety regulations overhauled.

5. Tropical Storms

Tropical storm (tropical cyclone): An intense low-pressure weather system with sustained wind speeds of 119+ km/h (called hurricanes in the Atlantic/NE Pacific, typhoons in the NW Pacific, and cyclones in the Indian Ocean/S Pacific). They form over warm tropical oceans (SST >26.5°C to a depth of 60 m) between 5° and 30° north and south of the equator, where the Coriolis effect is sufficient to create rotation.

Formation Conditions

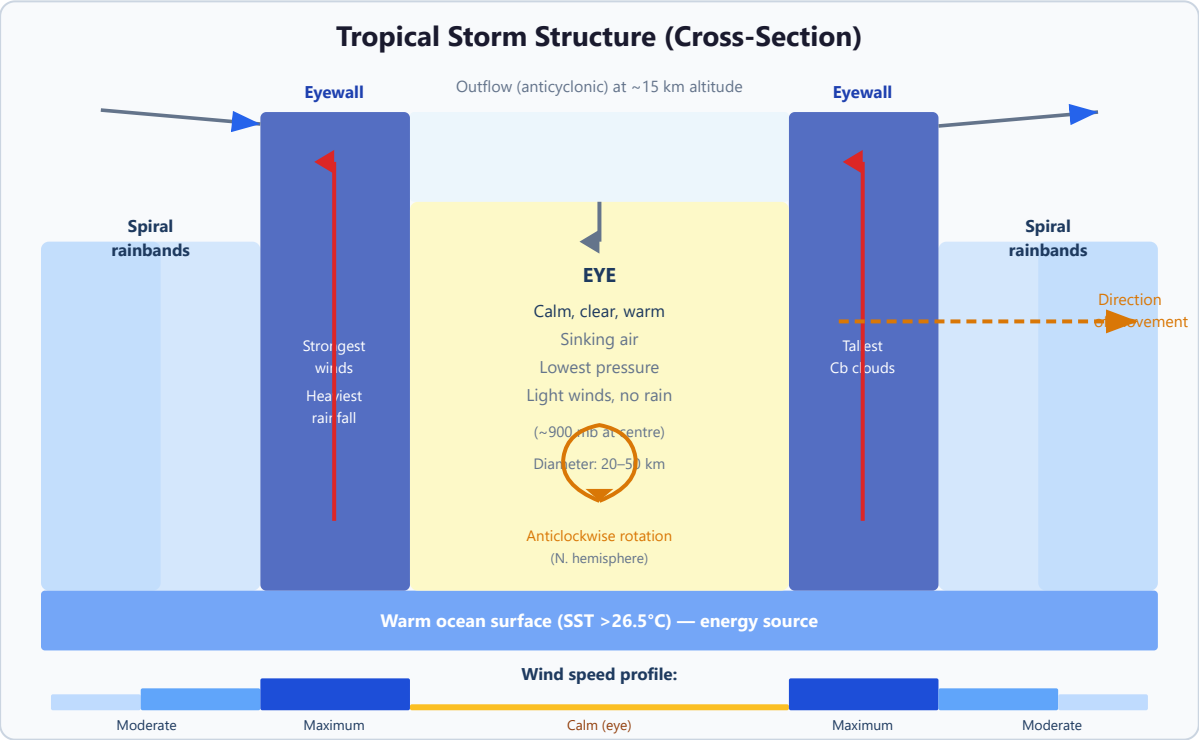
- Sea surface temperature >26.5°C (provides heat energy through evaporation)
- Deep warm water (at least 60 m depth)
- Location between 5°–30° latitude (Coriolis effect needed for spin)
- Low wind shear (uniform wind speed/direction at different altitudes allows the storm to build vertically)
- Pre-existing low-pressure disturbance (e.g. tropical wave, ITCZ)

- Moist, unstable atmosphere

Structure and Behaviour

A mature tropical cyclone has a distinct structure:

- **Eye** — calm, clear centre, 20–50 km diameter, with sinking air, light winds, and no rain. Air pressure lowest here (can drop to 900 mb or less in the strongest storms).
- **Eye wall** — ring of towering cumulonimbus clouds surrounding the eye. Most intense rainfall, strongest winds, and greatest uplift. This is where the storm is most destructive.
- **Spiral rain bands** — bands of cumulonimbus clouds extending outward from the eye wall, producing heavy rain and gusty winds. Can extend 500+ km from the centre.
- **Outflow** — at the top of the storm (~15 km altitude), air spirals outward, cooling and sinking to complete the circulation.



Tropical storms move westward in the trade wind belt at ~15–30 km/h, often curving poleward as they encounter the westerlies. They weaken when they move over cooler water, encounter strong wind shear, or make landfall (cut off from their energy source of warm ocean water).

Tropical Storm Hazards

- **Extreme winds** — sustained winds up to 300 km/h (Category 5). Destroy buildings, uproot trees, create deadly flying debris. Measured on the Saffir-Simpson Scale (1–5).
- **Storm surge** — a dome of seawater (up to 8 m above normal) pushed ashore by low pressure and onshore winds. The deadliest element of tropical storms. Cyclone Nargis (Myanmar, 2008) storm surge killed 138,000.
- **Intense rainfall and flooding** — can exceed 500 mm in 24 hours. Causes river flooding, flash floods, and landslides. Hurricane Harvey (2017) dumped 1,539 mm on Houston over 4 days.
- **Tornadoes** — some tropical storms spawn tornadoes in their outer rain bands.
- **Coastal erosion** — powerful waves and storm surge cause rapid coastal erosion.

Climate Change and Tropical Storms

Warmer sea surface temperatures (due to climate change) provide more energy for tropical storms. Scientific consensus: climate change is likely to increase the *intensity* of the strongest storms (more Category 4 and 5 events), increase rainfall rates (~7% more per 1°C of warming), and raise storm surge heights (due to sea-level rise). The *frequency* of storms may not increase overall, but the proportion of intense storms is expected to grow. The geographic range of tropical storms may expand poleward.

Case Study: Typhoon Haiyan (Yolanda), Philippines (November 2013)

Typhoon Haiyan was one of the most powerful tropical cyclones ever recorded, making landfall on 8 November 2013 with sustained winds of 315 km/h (Category 5) and gusts up to 380 km/h. Storm surge reached 5–7 m in Tacloban City. **Impacts:** 6,300 deaths (mostly from storm surge drowning), 28,689 injured, 4 million displaced, 1.1 million homes destroyed, ~$2.9 billion damage. Tacloban (population 220,000) was devastated — 90% of the city damaged or destroyed. Infrastructure (roads, bridges, power, water) was destroyed. **Response:** Philippine government declared State of National Calamity. International aid mobilised (US deployed aircraft carrier USS George Washington, UK sent HMS Illustrious). However, aid delivery was slow due to destroyed roads and airports. **Recovery:** $3.9 billion pledged by international donors. Resettlement of survivors was slow and controversial (many refused to move inland). 3 years later, 60,000+ families were still in temporary shelters. The disaster exposed extreme vulnerability: poverty (40% of affected area below poverty line), informal settlements on the coast, weak building construction, limited early warning reach in remote areas. The Philippines experiences ~20 typhoons per year due to its location in the western Pacific warm pool.

Case Study: Hurricane Katrina, USA (August 2005)

Katrina made landfall on the Gulf Coast as a Category 3 hurricane (sustained winds 205 km/h, storm surge up to 8.5 m). **Impacts:** 1,836 deaths, 80% of New Orleans flooded when levees breached, 1.3 million evacuated, 300,000 homes destroyed, $125 billion damage (costliest US natural disaster at the time). Disproportionate impact on poor, predominantly African American communities in the Lower 9th Ward, who lacked transport to evacuate and insurance to recover. **Response failures:** FEMA (Federal Emergency Management Agency) was widely criticised for slow and inadequate response. The Superdome emergency shelter was overcrowded and under-supplied. Pre-storm evacuation orders were issued but many could not comply. **Recovery:** $120+ billion in federal aid. New $14.5 billion hurricane protection system built (improved levees, floodwalls, pumping stations). Population of New Orleans fell from 484,000 (2000) to 343,000 (2010) — many residents never returned. Katrina exposed deep social inequalities: race, poverty, and institutional failure amplified the disaster far beyond what the storm's physical strength would suggest.

6. Fires in Nature

Wildfire: An uncontrolled fire burning through vegetation. Also called bushfires (Australia), forest fires, or wildfires. Wildfires are a natural part of many ecosystems but become hazardous when they threaten human life, property, and livelihoods. Climate change is increasing wildfire frequency and severity in many regions.

Conditions for Wildfires

- **Fuel** — dry vegetation (dead leaves, branches, grass, scrub, forest). Fuel load increases during drought. Eucalyptus trees in Australia contain flammable oils.
- **Weather** — high temperatures, low humidity, strong winds (spread fire rapidly and carry embers), prolonged drought.
- **Ignition** — lightning strikes (natural), human causes (arson, campfires, agricultural burning, power lines). ~85% of wildfires in the US are human-caused.
- **Topography** — fires spread faster uphill (preheating fuel above), through narrow valleys (chimney effect), and on slopes facing the sun.

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Climate Change and Wildfires

Rising temperatures, longer droughts, earlier snowmelt, and drier vegetation are increasing wildfire risk globally. Fire seasons are lengthening: in the western USA, fire season is now 78 days longer than in the 1970s. The area burned annually in California has increased fivefold since the 1970s.

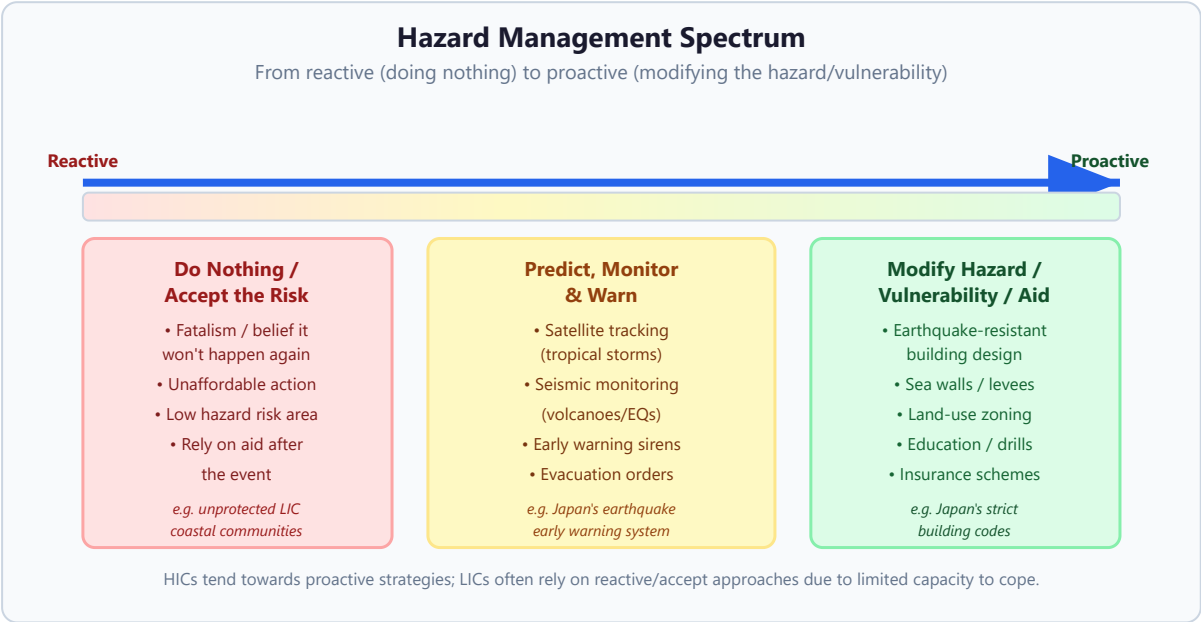
Case Study: Australian Black Summer Bushfires (2019–2020)
From September 2019 to March 2020, catastrophic bushfires burned across south-eastern Australia during record-breaking heat (national average temperature exceeded 41°C on 18 December 2019) and drought. **Impacts:** 34 direct deaths, 417 deaths from smoke inhalation, ~18.6 million hectares burned (an area larger than England and Wales combined), 5,900 buildings destroyed (including 2,779 homes), 3 billion animals killed or displaced (including an estimated 1 billion mammals, birds, and reptiles), air quality in Sydney reached 11 times the hazardous threshold. Economic cost: ~A$100 billion. **Cause:** combination of extreme drought (worst on record in many areas), record temperatures (linked to Indian Ocean Dipole and anthropogenic climate change), low humidity, and strong winds. Natural lightning ignited many fires. **Response:** largest peacetime military deployment in Australian history, international firefighters from US, Canada, and NZ. **Significance:** the fires generated public pressure for stronger climate action, highlighted the increasing severity of wildfire hazards under climate change, and demonstrated that even wealthy countries with experienced fire services can be overwhelmed.

7. Multi-Hazard Environments

Multi-hazard environment: A location exposed to two or more types of natural hazard. Many of the world's most hazardous areas face multiple threats simultaneously (e.g. earthquakes, volcanic eruptions, tsunamis, tropical storms, flooding, landslides).

Case Study: The Philippines — A Multi-Hazard Environment
The Philippines (population ~115 million, 7,641 islands) is one of the world's most hazard-prone countries, facing: **Earthquakes** (located on the Pacific Ring of Fire, Philippine Fault zone, destructive boundary); **Volcanic eruptions** (22 active volcanoes, including Taal and Mayon); **Tropical storms** (~20 typhoons/yr cross the Philippines, the most of any country); **Flooding and landslides** (monsoon rainfall, deforestation, steep terrain); **Tsunamis** (1976 Moro Gulf tsunami killed 5,000). Between 2000

and 2019, the Philippines experienced 315 recorded disasters, affecting 130 million people. Vulnerability factors: high poverty rates (16.7% below poverty line), rapid urbanisation, coastal settlement, informal housing, deforestation increasing landslide risk. Resilience building: Philippines Disaster Risk Reduction and Management Act (2010), improved early warning systems, community-based disaster preparedness programmes, building code reforms.



Exam Tip: AQA 20-mark hazard essays often require you to compare and evaluate. For example: "To what extent is the level of development the main factor affecting the impact of tectonic hazards?" Structure: (1) argue FOR development as the key factor (building quality, emergency services, insurance, governance), (2) argue AGAINST (magnitude, depth, population density, time of day, secondary hazards also matter), (3) evaluate and reach a judgement. Use paired case studies (HIC vs LIC) with specific data to support your argument.

Exam Tip: Always distinguish between *prediction* (forecasting when/where a hazard will occur), *protection* (engineering measures to reduce damage), and *preparedness* (planning and education). Earthquakes cannot be reliably predicted, but volcanic eruptions can be monitored for warning signs (seismicity, gas emissions, ground deformation). Tropical storms can be tracked with satellite imagery, giving days of warning. Discuss what level of warning is possible for each hazard type.

Exam Tip: The concept of the **disaster-development nexus** is important at AQA. Disasters can set back development (Nepal 2015 lost 50% of GDP), while lack of development increases vulnerability to disasters. This creates a vicious cycle. Conversely, disasters can trigger development improvements ("build back better"). Use this framework when discussing hazard impacts in LICs.